UNIT 2 · TOPIC 2.4

Physical Detection Systems

Quiz

Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. Which of the following is a detective physical security control?

- ○ A. A deadbolt lock on a server room door
- ○ B. Full-disk encryption on all workstations
- ○ C. A motion-activated CCTV camera inside the server room
- ○ D. A multi-factor authentication system on network accounts

2. A badge reader at the entrance of Xtensr Labs records every access attempt, including attempts where the card is rejected. What type of security control does this logging capability represent?

- ○ A. Preventive
- ○ B. Deterrent
- ○ C. Corrective
- ○ D. Detective

3. Which of the following best describes the purpose of a tamper-evident seal placed on a server chassis?

- ○ A. It prevents unauthorized users from logging in to the server remotely.
- ○ B. It provides a visible indication that the chassis may have been physically opened or altered.
- ○ C. It encrypts the data stored on internal drives.
- ○ D. It automatically alerts the network operations center when removed.

4. While reviewing access logs for Xtensr Labs' new facility, a security analyst notices the same employee badge was used at the main entrance and at a rear server room 18 seconds apart, a distance that takes at least 4 minutes to walk. What detection technique is the analyst applying, and what does the finding suggest?

- ○ A. Perimeter monitoring; the badge reader hardware is malfunctioning and timestamps are unreliable.
- ○ B. Network traffic analysis; a replay attack is being launched against the access control system.
- ○ C. Biometric analysis; the employee's identity could not be verified at one of the readers.
- ○ D. Access log anomaly detection; the badge may be cloned or shared, indicating a potential physical security breach.

5. Xtensr Labs installs pressure sensors beneath the floor tiles of its server room. A security engineer explains that the sensors will trigger an alert if weight is detected in the room when no badge access has been logged. What type of physical attack is this detection method specifically designed to identify?

- ○ A. Unauthorized physical intrusion through a path that bypasses the badge reader
- ○ B. Eavesdropping on wireless signals emitted by the servers
- ○ C. Denial-of-service attacks targeting the servers over the network
- ○ D. Social engineering of an employee in a common area

6. A security team wants evidence that can support a forensic investigation if a physical breach occurs. Which combination of controls best provides that evidence?

- ○ A. Timestamped CCTV recording with 90-day retention and badge access logs that capture entry, exit, and failed attempts
- ○ B. Perimeter fencing and a roving security guard
- ○ C. Biometric entry scanner and mantrap vestibule
- ○ D. Cable locks on servers and privacy screen filters on monitors

7. During a physical vulnerability assessment of Xtensr Labs' newly acquired facility, you find: a badge reader that logs only successful entries, a CCTV camera covering the front entrance only, and no intrusion alarm system. Which gap in this detection setup creates the most significant risk to post-incident forensic analysis?

- ○ A. The failure to log failed access attempts alongside successful ones
- ○ B. The absence of perimeter fencing around the building
- ○ C. The lack of cable locks on server hardware
- ○ D. No documented visitor badge policy

8. When evaluating the effectiveness of a physical monitoring system for a research lab, which metric most accurately measures how well the system identifies real intrusion events without generating excessive noise that causes security staff to ignore alerts?

- ○ A. Total number of sensors installed per square meter
- ○ B. The system's detection rate measured against its false alarm rate
- ○ C. The average response time of the on-call security officer
- ○ D. Per-unit cost of each installed sensor

9. A physical security analyst discovers that motion sensors in two adjacent server corridors have overlapping zones, yet an attacker who moves sufficiently slowly can pass through both zones without triggering either sensor, because each sensor's movement threshold is set for typical human walking speed. What class of vulnerability does this represent, and which enhancement best corrects it?

- ○ A. Sensor redundancy conflict; corrected by removing one sensor to eliminate zone overlap
- ○ B. Insufficient ambient lighting; corrected by installing brighter LED arrays in the corridors
- ○ C. Network misconfiguration in the sensor management console; corrected by applying the latest firmware patch
- ○ D. Movement-speed threshold blind spot in motion detection; corrected by supplementing with passive infrared (PIR) heat-signature sensors and implementing cross-sensor event correlation

10. Which security strategy uses multiple overlapping detection layers so that if one layer fails to identify a threat, another layer remains in place to catch it?

- ○ A. Least privilege
- ○ B. Zero-day protection
- ○ C. Defense in depth
- ○ D. Patch management

11. A retail facility installs clearly visible CCTV cameras at every entrance and posts signs reading "This area is under 24-hour video surveillance." Beyond its detection function, what additional security role do these visible cameras serve?

- ○ A. Deterrence of potential attackers
- ○ B. Data encryption of video feeds

- ○ C. Network segmentation of the camera system
- ○ D. Active prevention of physical intrusion

12. Which of the following is NOT a physical detection control?

- ○ A. A passive infrared motion sensor
- ○ B. A door contact alarm that triggers when the door is opened
- ○ C. A CCTV camera with continuous recording
- ○ D. A reinforced steel deadbolt lock

13. Xtensr Labs is designing camera placement for a server room that has a single badge-controlled entrance. The security goal is to detect both tailgating (a second person entering immediately behind an authorized user) and unauthorized removal of equipment from the racks. Which camera placement strategy best achieves both goals?

- ○ A. One camera mounted inside the room pointing at the server racks
- ○ B. One camera facing inward at the entrance to capture all entrants, and one camera covering the server rack area
- ○ C. One camera mounted outside pointing at the entrance door
- ○ D. Cameras mounted only at exterior building perimeter entrances

14. A physical security team is selecting a motion sensor for a storage area that houses active cooling equipment generating significant ambient heat. Which sensor type is most appropriate, and why?

- ○ A. Passive infrared (PIR) sensor, because it is the least expensive option
- ○ B. PIR sensor, because it provides a wider coverage angle than microwave sensors
- ○ C. Microwave motion sensor, because it detects movement based on reflected signal change rather than heat signatures
- ○ D. Either type performs equally well; both sensors detect identical signals

15. An organization needs to detect if server hardware is physically removed from its rack without authorization. Which detection strategy most directly addresses this threat?

- ○ A. Installing weight sensors in the equipment racks integrated with an asset management alert system
- ○ B. Encrypting the hard drives within each server
- ○ C. Requiring two-factor authentication for server room badge access
- ○ D. Scheduling weekly manual equipment inventory audits

16. A physical security review of Xtensr Labs' newly acquired facility reveals: badge readers at the main entrance only, CCTV cameras covering public corridors but not the server or lab areas, and motion sensors that are disabled each evening to prevent false alarms triggered by the cleaning crew. Evaluate the most critical flaw in this detection strategy from a threat exposure standpoint.

- ○ A. Disabling motion sensors after business hours creates a detection gap during the period when the facility is most vulnerable to undetected physical intrusion.
- ○ B. Badge readers at the main entrance should be upgraded to biometric scanners to reduce credential theft risk.
- ○ C. CCTV cameras in public corridors are redundant with badge readers and should be reallocated to server areas.
- ○ D. The facility should replace all electronic monitoring with dedicated on-site security personnel.

17. NIST SP 800-53 control PE-6 requires organizations to monitor physical access to facilities. A security architect must design a detection strategy for a five-floor research building where the top two floors contain restricted research labs and the lower three floors contain general office space. Which approach best satisfies PE-6 while maintaining operational efficiency and proportionality?

- ○ A. Apply identical detection controls to all five floors regardless of sensitivity level to ensure consistent coverage.

○ B. Apply tiered detection controls scaled to asset sensitivity: restricted lab floors receive CCTV, badge access logs, and motion sensors; general office floors receive CCTV coverage only.

○ C. Assign a dedicated security officer to each floor to eliminate the need for electronic monitoring.

○ D. Monitor only the building perimeter and rely on internal access controls to protect sensitive areas.

18. A workstation is stolen from Xtensr Labs over a weekend. A post-incident review finds: CCTV footage from the affected area is corrupted due to a storage failure, badge access logs were overwritten after 48 hours as configured, and motion sensor alerts were sent to a single email address belonging to a staff member who was on leave. Which systemic failure in the detection strategy does this incident most clearly expose, and what is the highest-priority remediation?

○ A. The CCTV cameras were insufficient resolution; replace with high-definition units to ensure footage quality.

○ B. Motion sensors should be replaced with pressure-plate sensors for greater accuracy.

○ C. The facility needed more security guards patrolling on weekends.

○ D. The detection system lacked redundancy in alert routing and log retention; highest-priority remediation is implementing multi-channel alert escalation and extending access log retention to meet incident investigation timelines.